Yixuan Zhang

Contact

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Education

- **University of Wisconsin–Madison** — Madison, WI
  Ph.D. in Industrial & Systems Engineering — 2022–May 2027 (expected)
  – Advisor: Prof. Qiaomin Xie
  M.S. in Computer Science — 2022–2023
  M.A. in Mathematics — 2021–2022
- **Nankai University** — Tianjin, China
  B.S. in Applied Mathematics — 2017–2021

Research Interests

My research develops mathematical foundations for machine learning and sequential decision-making by combining applied probability with algorithm design. My current work centers on two directions:

1. **Distributional analysis of stochastic algorithms.** I establish distributional convergence and characterize the long-run distributions and bias of stochastic algorithms, using these results to guide algorithmic improvements.
2. **Satisficing sequential decision-making under imperfect information.** I develop algorithms that maintain satisfactory performance while avoiding unnecessary learning costs under imperfect information, including when the available historical data are biased or the environment is nonstationary.

Publications

Journal Publications

[J1] Prelimit Coupling and Steady-State Convergence of Constant-Stepsize Nonsmooth Contractive Stochastic Approximation
Yixuan Zhang, Dongyan Huo, Yudong Chen, and Qiaomin Xie.
Operations Research. Accepted for publication.

Conference Publications

[C1] Wasserstein-p Central Limit Theorem Rates: From Local Dependence to Markov Chains
Yixuan Zhang and Qiaomin Xie.
ACM SIGMETRICS, 2026.
– **Best Student Paper Award (Best Paper Finalist), ACM SIGMETRICS 2026.**

[C2] Contextual Online Pricing with (Biased) Offline Data
Yixuan Zhang, Ruihao Zhu, and Qiaomin Xie.
Advances in Neural Information Processing Systems (NeurIPS), 2025.

[C3] A Piecewise Lyapunov Analysis of Sub-quadratic SGD: Applications to Robust and Quantile Regression
Yixuan Zhang, Dongyan Huo, Yudong Chen, and Qiaomin Xie.
ACM SIGMETRICS, 2025.

[C4] The Collusion of Memory and Nonlinearity in Stochastic Approximation with Constant Stepsize
Dongyan Huo, **Yixuan Zhang**, Yudong Chen, and Qiaomin Xie.
Advances in Neural Information Processing Systems (NeurIPS), 2024. **Spotlight.**

[C5] Prelimit Coupling and Steady-State Convergence of Constant-Stepsize Nonsmooth Contractive Stochastic Approximation
Yixuan Zhang, Dongyan Huo, Yudong Chen, and Qiaomin Xie.
ACM SIGMETRICS, 2024.

[C6] Constant Stepsize Q-learning: Distributional Convergence, Bias and Extrapolation
Yixuan Zhang and Qiaomin Xie.
Reinforcement Learning Conference (RLC), 2024.

Journal Manuscripts under Revision

[R1] A Unified Lyapunov Analysis for Sub-quadratic SGD: Applications to Robust and Quantile Regression
Yixuan Zhang, Dongyan Huo, and Qiaomin Xie.
– Major revision, *Stochastic Systems*.

[R2] Satisficing Regret Minimization under Nonstationarity
Yixuan Zhang, Ruihao Zhu, and Qiaomin Xie.
– Major revision, *Operations Research*.
– **Finalist, INFORMS 2026 Data Mining Society Best Student Paper Award Competition.**

[R3] Memory, Nonlinearity, and Drift–Contraction in Stochastic Approximation with a Constant Stepsize
Yixuan Zhang, Dongyan Huo, and Qiaomin Xie.
– Major revision, *Mathematics of Operations Research*.

Working Papers

[W1] Steady-State Convergence of Stochastic Approximation
Yixuan Zhang and Qiaomin Xie.
– *To be submitted to Operations Research.*
– **First Place, ACM SIGMETRICS 2026 Student Research Competition.**

[W2] Gaussian Approximation in Wasserstein-p: Markovian Martingale Differences and Stochastic Approximation
Yixuan Zhang and Qiaomin Xie.
– *To be submitted to Mathematics of Operations Research.*

[W3] Gaussian Approximation in Wasserstein-1: Local Dependence and Markovian Martingale Differences
Yixuan Zhang and Qiaomin Xie.
– *To be submitted to Mathematics of Operations Research.*

Honors and Distinctions

- **Best Student Paper Award** (**Best Paper Finalist**), ACM SIGMETRICS 2026
- **First Place**, ACM SIGMETRICS Student Research Competition 2026
- **Finalist**, INFORMS Data Mining Society Best Student Paper Award Competition 2026
- Finalist, NUS Next-Gen Scholar Symposium 2026
- Narayan-Chen Family Graduate Student Support Scholarship, University of Wisconsin–Madison 2026
- OM Rookiepalooza, Northwestern University 2026
- ISyE–MS&E–IOE Joint Rising Stars, Georgia Tech 2025

Research Mentorship

- **Yanbo Qiao** (Ph.D. student in ISyE, UW–Madison) Sep. 2026–Present
  – *Finite-Time Analysis of Quantile TD Learning*
- **Yuqi Zeng** (Undergraduate student, CUHK–Shenzhen) Jan. 2026–Present
  – *Quotient-Space Distributional Analysis of Average-Reward TD Learning*
- **Yining Liu** (M.S. student in Mathematics, UW–Madison) Dec. 2025–Mar. 2026
  – *Distributional Analysis of Asynchronous SGD: When Delays Do Not Hurt*

Teaching

- **Teaching Assistant, University of Wisconsin–Madison**
  – ISyE 624: Stochastic Modeling Techniques Spring 2023 and 2025
  – ISyE 313: Engineering Economic Analysis Fall 2022
  – ISyE 525: Linear Optimization Summer 2021
- **Guest Lecturer, University of Wisconsin–Madison**
  – ISyE 624: Stochastic Modeling Techniques Spring 2023 and 2025

Tutorial

- **Distributional Analysis of Stochastic Algorithms** 2025
  – ACM SIGMETRICS; 90-minute tutorial.

Selected Talks

- **Gaussian Approximation in Wasserstein-p: Markovian Martingale Differences and Stochastic Approximation**
  - INFORMS Annual Meeting, San Francisco 2026
- **Satisficing Regret Minimization under Nonstationarity**
  - INFORMS Annual Meeting (Award Session), San Francisco 2026
  - INFORMS Optimization Society Conference, Atlanta 2026
- **Steady-State Convergence of Stochastic Approximation**
  - Analytics for X, National University of Singapore 2026
- **Wasserstein-p Central Limit Theorem Rates: From Local Dependence to Markov Chains**
  - ACM SIGMETRICS, University of Michigan 2026
  - OM Rookiepalooza, Northwestern University 2026
  - NYC Operations Day PhD Colloquium 2026
- **Contextual Online Pricing with (Biased) Offline Data**
  - INFORMS Annual Meeting, Atlanta 2025
- **A Piecewise Lyapunov Analysis of Sub-quadratic SGD: Applications to Robust and Quantile Regression**
  - INFORMS Applied Probability Society Conference, Atlanta 2025
  - ACM SIGMETRICS, Stony Brook University 2025
- **Prelimit Coupling and Steady-State Convergence of Constant-stepsize Nonsmooth Contractive SA**
  - ISyE–MS&E–IOE Joint Rising Stars Workshop, Georgia Tech 2025
  - INFORMS Annual Meeting, Seattle 2024
  - ACM SIGMETRICS, Italy 2024

Service

- **Session Chair**
  - INFORMS Annual Meeting 2025 and 2026
  - INFORMS Applied Probability Society Conference 2025
- **Refereeing**
  - *Journals:* IEEE Transactions on Information Theory; Mathematics of Operations Research; Physica D: Nonlinear Phenomena.
  - *Conferences:* NeurIPS; Reinforcement Learning Conference (RLC).

References

- **Qiaomin Xie** (Assistant Professor; Ph.D. Advisor)
  - Department of Industrial and Systems Engineering, University of Wisconsin–Madison
  - Email: qiaomin.xie@wisc.edu
- **Yudong Chen** (Professor)
  - Department of Computer Sciences, University of Wisconsin–Madison
  - Email: yudongchen@cs.wisc.edu
- **Ruihao Zhu** (Assistant Professor)
  - Cornell SC Johnson College of Business
  - Email: ruihao.zhu@cornell.edu